

AAMA 502-21

An FGIA Specification

Voluntary Specification for Field Testing of Newly Installed Fenestration Products

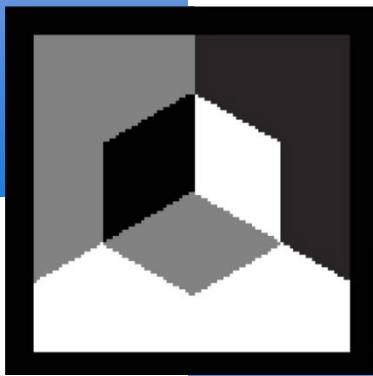


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HOW TO USE THE SHORT FORM SPECIFICATION

To standardize and simplify the writing of field-testing specifications for fenestration products, FGIA has prepared the following "Short Form Specification" for use by architects/specifiers. It may be used by merely inserting the following paragraph(s) into the project specifications. Please note that Specifier Notes in italics are not to be inserted into the project specifications.

SHORT FORM FIELD TESTING SPECIFICATION

1. Newly installed fenestration product(s) shall be field tested, as contractually agreed upon by the interested parties, in accordance with AAMA 502, "Voluntary Specification for Field Testing of Newly Installed Fenestration Products" by a laboratory or test agency accredited by AAMA to this specification.

***SPECIFIER NOTE 1:** See Section 5.0 for a description of the test method. AAMA 502 specifies air leakage resistance and water penetration resistance field testing of fenestration product specimens during construction and prior to issuance of the building occupancy permit.*

2. Unless otherwise specified, test 1% of the fenestration product specimens, but no fewer than three, after those products have been completely installed for air leakage resistance and water penetration resistance as specified.

***SPECIFIER NOTE 2:** The number of specimens to be selected for testing on a project should be determined after careful consideration of the following factors:*

- Selecting a large number of specimens on a small project is normally not cost effective.*
- On all projects, regardless of size, the specifier should determine how many specimens are required in order to establish a reasonable measure of quality for the entire project. ASTM E122 provides guidance on how to establish the number of test specimens in order to estimate a measure of quality of a production lot with prescribed precision.*
- The specifier should carefully balance the cost of specimen or specimen preparation, as well as the cost of testing, and the restoration of the project surrounding components and finishes to their original condition, before determining the number and location of specimens to be tested. On larger projects, a formal cost-benefit analysis is appropriate. On smaller projects, the specifier should avoid a situation where the cost of testing and building restoration approaches the cost of the fenestration products. If any of the tested fenestration specimens fail to comply with the project requirements, consideration should be given to the selection and testing of additional products. (See AAMA 502 Section 7.5.9.)*
- If water penetration is observed and the source of the penetration cannot be determined, a forensic evaluation using the procedures outlined in AAMA 511 shall be performed while maintaining the test pressures and methods defined in AAMA 502.*

3. Air leakage resistance tests shall be conducted at a uniform static test pressure of ____ Pa (____ psf). The maximum allowable rate of air leakage shall not exceed ____ L/s•m² (____ cfm/ft²).

***SPECIFIER NOTE 3:** The field test pressure for air leakage resistance shall be equivalent to the test pressure originally used in the laboratory to test the product or achieve a laboratory performance rating. Unless stated otherwise and clearly stipulated in the project specifications, the allowable rates for field air leakage testing shall be a minimum of 1.5 times the applicable AAMA/WDMA/CSA 101/I.S.2/A440 rate for the Product Type and Performance Class, rounded to one decimal place.*

4. Water penetration resistance tests shall be conducted at a static test pressure of ____ Pa (____ psf). No water penetration shall occur as defined in AAMA 502 Section 7.5.8.1.

***SPECIFIER NOTE 4:** Unless stated otherwise and clearly stipulated in the project specifications, in no case shall the specified static water test pressure differential exceed 2/3 of the tested or rated laboratory performance (see AAMA 502 Section 7.5.3 for default performance requirements). If any window fails to meet the requirements, refer to AAMA 502 Section 7.5.9.*

0.0 INTRODUCTION

0.1 The purpose of this specification is to provide a method to evaluate the installed performance of newly installed fenestration products for air leakage resistance and water penetration resistance under controllable, reproducible and appropriate conditions for the purpose of providing reasonable assurance that as installed, the fenestration products will perform to the specified or rated air leakage and water penetration performance levels with allowances given for field performance.

0.2 This specification establishes the requirements for field test specimens, apparatus, sampling, test procedures and test reports to be used in verifying the air infiltration resistance performance and water penetration resistance performance of newly installed fenestration products.

1.0 SCOPE

1.1 This specification is applicable during construction, prior to issuance of the building occupancy permit but no later than six (6) months after installation of the fenestration product. For the purpose of this document, fenestration products are defined in AAMA/WDMA/CSA 101/I.S.2/A440. Specifically excluded are curtain walls, sloped glazing and storefront systems; these systems shall be tested per AAMA 503 as required. Verifying the structural loading resistance performance of fenestration products is beyond the scope of this specification.

1.2 If field testing of fenestration products is required after the building occupancy permit has been issued or more than six (6) months after product installation, AAMA 511 shall be used. Additionally, in the event where water penetration testing using the test methods outlined in this specification produces water infiltration that cannot be solely attributed to the fenestration product, forensic testing shall be initiated by using AAMA 511 and ASTM E2128. The determination of the sources of water penetration from other than the fenestration product requires forensic investigation and testing outside the scope of this specification.

1.3 Terminology

In this specification, “shall” is used to express a requirement, i.e., a provision that the user is obliged to satisfy in order to comply with the specification; “shall be permitted to be” is used to express an option or that which is permissible within the limits of the specification; “should” is used to express a recommendation or that which is advised but not required; and “may” is used to express possibility or capability. Notes accompanying sections do not include requirements or alternative requirements; the purpose of a note accompanying a section is to separate explanatory or informative material from the text. Notes to tables and figures are considered part of the table or figure and shall be permitted to be written as requirements.

1.4 The primary units of measure in this document are metric. The values stated in SI units are to be regarded as the standard. The values given in parentheses are for reference only.

1.5 This document was developed in an open and consensus process and is maintained by representative members of FGIA as advisory information.

2.0 SIGNIFICANCE AND USE

2.1 This document provides a method which shall be used to evaluate the performance of newly installed fenestration products for air leakage rate and water penetration resistance under controllable, reproducible and appropriate conditions.

2.2 The specifier/architect shall use this document and the short form provided to communicate to the fenestration product manufacturer the installed performance for air infiltration and water penetration. Following product installation, this document provides a structured, controllable and reproducible method for testing.

2.3 Improper use of this document may lead to a misunderstanding of a fenestration product’s performance and/or the misapplication of fenestration products on the project.

2.4 The use of this document for the testing of installed fenestration products presumes that all personnel comply with the requirements of AAMA LAP-3, *Laboratory Accreditation Program Operations Manual - Laboratories and Test Agencies Performing On-site Testing of Fenestration Products*. These requirements include, but are not limited to, a working knowledge of the following:

2.4.1 Building construction and window, door and glazed wall section installation techniques.

2.4.2 The various window types and designs, their operation and application.

2.4.3 Understanding how the placement and attachment of air and water test chambers can affect test results.

2.4.4 The understanding of how to isolate leakage through system components such as window, trim/panning, perimeter and other extraneous leakage.

2.4.5 How interior and exterior ambient conditions (temperature, wind, barometric pressure, and moisture) affect measurement accuracy and how they affect the specimen being tested.

2.4.6 The principles of pressure measurement, fluid mechanics and instrumentation practices.

2.4.7 Assurance that on-site verifications are conducted in accordance with Section 8.0.

2.5 All testing shall be conducted by an AAMA accredited laboratory or field test agency. See AAMA LAP-3 for more details.

3.0 REFERENCED DOCUMENTS

3.1 References to the standards listed in this section shall be to the edition indicated. Any undated reference to a code or standard appearing in the requirements of this specification shall be interpreted as referring to the latest edition of that code or standard.

3.2 AAMA, Fenestration and Glazing Industry Alliance (FGIA) Standards

AAMA/WDMA/CSA 101/I.S.2/A440-05, Standard Specification for Windows, Doors, and Unit Skylights

AAMA/WDMA/CSA 101/I.S.2/A440-08, North American Fenestration Standard/Specification for windows, doors, and skylights

AAMA/WDMA/CSA 101/I.S.2/A440-11, North American Fenestration Standard/Specification for windows, doors, and skylights

AAMA/WDMA/CSA 101/I.S.2/A440-17, North American Fenestration Standard/Specification for windows, doors, and skylights

AAMA LAP-3, Laboratory Accreditation Program Operations Manual - Laboratories and Test Agencies Performing On-site Testing of Fenestration Products

AAMA 503-14, Voluntary Specification for Field Testing of Newly Installed Storefronts, Curtain Walls and Sloped Glazing Systems

AAMA 511-08, Voluntary Guideline for Forensic Water Penetration Testing of Fenestration Products

3.3 ASTM International (ASTM)

ASTM E122-17, Standard Practice for Calculating Sample Size to Estimate, With Specified Precision, the Average for a Characteristic of a Lot or Process

ASTM E283/E283M-19, Standard Test Method for Determining Rate of Air Leakage Through Exterior Windows, Skylights, Curtain Walls, and Doors Under Specified Pressure Differences Across the Specimen

ASTM E783-02(2018), Standard Test Method for Field Measurement of Air Leakage Through Installed Exterior Windows and Doors

ASTM E1105-15, Standard Test Method for Field Determination of Water Penetration of Installed Exterior Windows, Skylights, Doors, and Curtain Walls, by Uniform or Cyclic Static Air Pressure Difference

ASTM E2128-17, Standard Guide for Evaluating Water Leakage of Building Walls

4.0 DEFINITIONS

Please refer to the most current AAMA Glossary for all definitions.

5.0 TEST METHODS

5.1 The field procedure and test apparatus shall meet the requirements of ASTM E783 and ASTM E1105. The AAMA accredited test agency shall make these test methods along with AAMA 502 available for review upon request.

5.2 Test Chamber Arrangement

The test chamber shall be attached and sealed to the wall/roof construction in such a manner as to create a uniform pressure differential across the entire specimen (including subframe/receptor and/or panning) and the perimeter seals (see Figure 1).

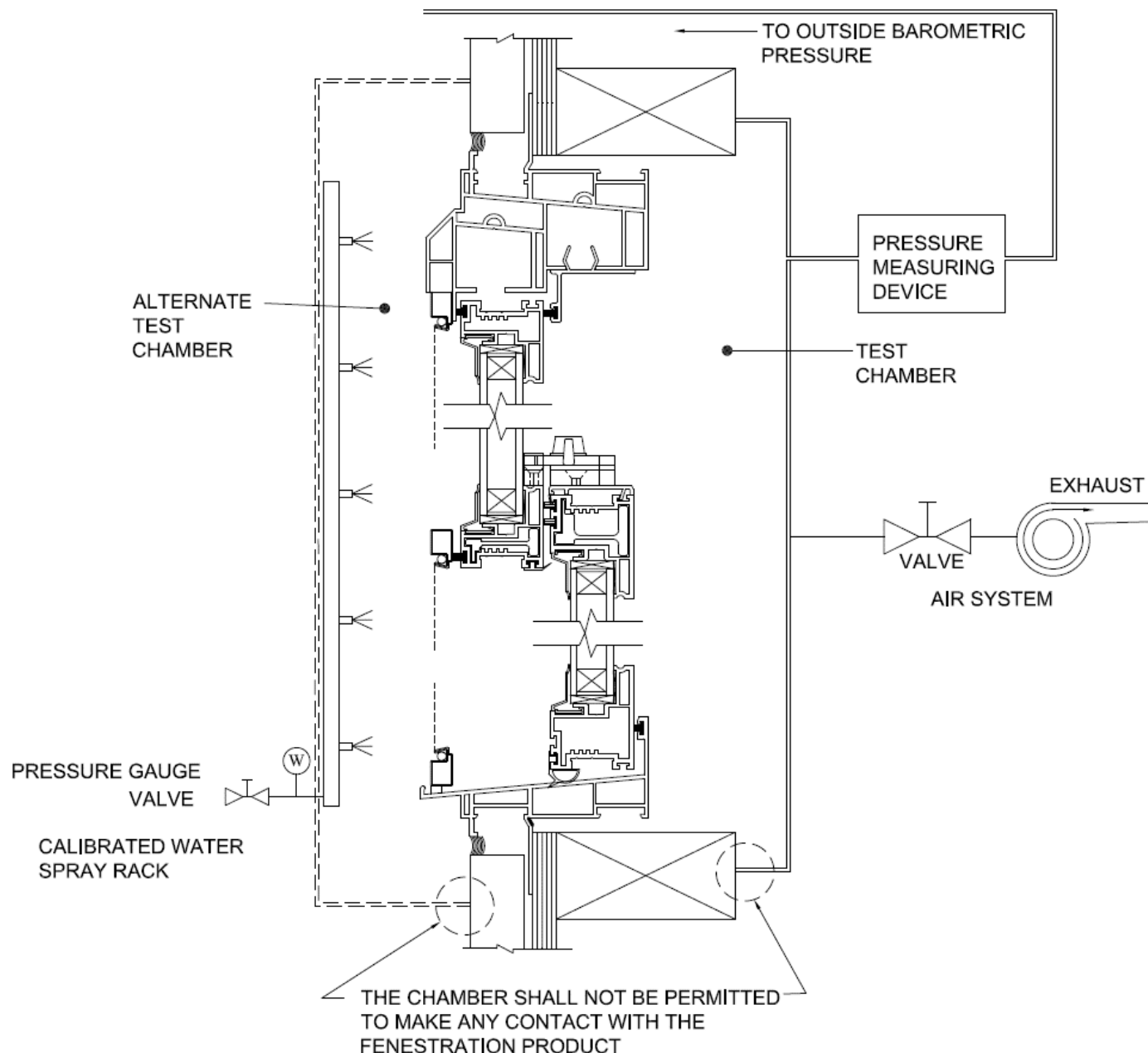


FIGURE 1: Test Chamber

NOTE 1: Wind and other environmental conditions (barometric pressure, rain and temperature changes) can have an adverse impact on field testing performance; therefore, the specimen should be isolated from adverse ambient conditions with a barrier and conditioned if necessary, to protect it from any environmental condition which may affect the test results. If testing must be performed under conditions that would affect the test results, such as extreme temperatures, high winds or other adverse environmental conditions, these conditions should be noted in the report. Any test performed without appropriate consideration and adjustment for ambient conditions may render the test results invalid.

5.2.1 Testing shall be performed as soon as possible after the fenestration product(s) is installed, and prior to the installation of drywall or interior finish materials. If interior finish materials have been installed, they shall be removed at the test area to allow visual access to these areas to check for water penetration, or other means of visual access shall be provided.

NOTE 2: The amount of interior finish materials to be removed should be agreed upon prior to testing. The cost of the repair and replacement of any interior finish materials removed should also be considered.

5.2.2 The test chamber shall be applied to the wall/roof system in such a manner as to apply a pressure differential to all joinery conditions and minimize extraneous air leakage (see Section 7.4.2 for a detailed description of extraneous air leakage).

5.2.2.1 In the case of a stud wall/roof system, the test chamber perimeter line, if installed on the building interior, would normally be attached to studs on both sides of the specimen and at the head and sill framing.

5.2.2.2 On a precast wall/roof system, a chamber installed on the interior can be sealed directly to the back of the precast panel.

5.2.2.3 Special sealing consideration shall be given to the testing of continuous strip windows. Tests on a segment of continuous horizontal strip fenestration products shall include adequate measures to properly seal horizontal head receptors and sill starters so that extraneous air leakage and water control is maintained.

6.0 SAMPLING, SITE PREPARATION AND RESPONSIBLE PARTIES

6.1 As soon as practical after installation has begun, and a representative number of fenestration products have been completely installed, adjusted, cleaned and perimeter sealed, the agreed upon number of installed fenestration product specimens shall be tested as specified in Section 7.0 for air leakage rate and/or water penetration resistance.

NOTE 3: Performing the field test as soon as practical may be beneficial in determining if manufacturing, installation and/or perimeter sealing problems are present before a substantial portion of the project is completed. If the initial testing is performed early in the project installation process, necessary corrections can be made without significant financial impact on any of the responsible parties. On large projects, tighter construction monitoring may be performed by testing at approximate intervals of 5%, 50% and 90% completion of the installation. Three field test specimens, as specified previously, is the default quantity. If more than, or less than, three specimens are to be tested, the number of specimens and intervals of testing shall be stipulated in Paragraph 2 of the Short Form Field Testing Specification.

6.2 The project specifications and/or the contract documents shall clearly identify the party responsible for the costs associated with the initial field testing, retesting and forensic testing (if necessary).

6.3 Testing shall be performed by an AAMA accredited independent laboratory or field test agency.

NOTE 4: AAMA accredited laboratories are required to comply with AAMA LAP-3, "Laboratory Accreditation Program Operations Manual - Laboratories and Test Agencies Performing On-site Testing of Fenestration Products." The requirement of AAMA accreditation assures the specifier that the laboratory has the staff, training, experience and calibrated equipment to properly perform field testing.

6.3.1 The entity or party contracting the field testing shall notify the fenestration product installer, manufacturer, and the responsible party defined in Section 6.2 of the test schedule. The initial advance notice shall be a minimum of two weeks in advance of the testing in order for the general contractor to notify all fenestration product trades (i.e., erector, glazier, perimeter caulk contractor, etc.) of the scheduled test date.

6.4 Fenestration product specimen(s) selected for testing shall conform to the following:

6.4.1 Specimen location shall be selected by the architect or owner's representative unless delegated to the AAMA accredited independent testing agency or building envelope consultant. If any of the specimens do not conform to the prescribed air leakage resistance and/or water penetration resistance requirements, the manufacturer and/or the installer shall be afforded the opportunity to perform a site inspection and determine the reason for non-compliance.

6.4.2 The fenestration product(s) shall be representative specimens of typical installations as specified for the project. All exterior and interior perimeter seals shown in the installation instructions shall be in place. The specimen(s) shall have no outstanding punch list items, visible damage or irregularities including but not limited to torn or damaged gaskets, bent frames, nor be singled out because of obvious performance problems. If problems with a specimen installation are observed, they shall be brought to the contractor's and/or manufacturer representative's attention and added to the project punch list. If exterior screens are specified, they shall be in place (closed) during testing.

6.4.3 Exterior cladding, rain screen and associated components, including the exterior perimeter sealants, shall be installed adjacent to the fenestration product to represent the final detailed conditions of the exterior envelope. Figure 2 provides an example of this.

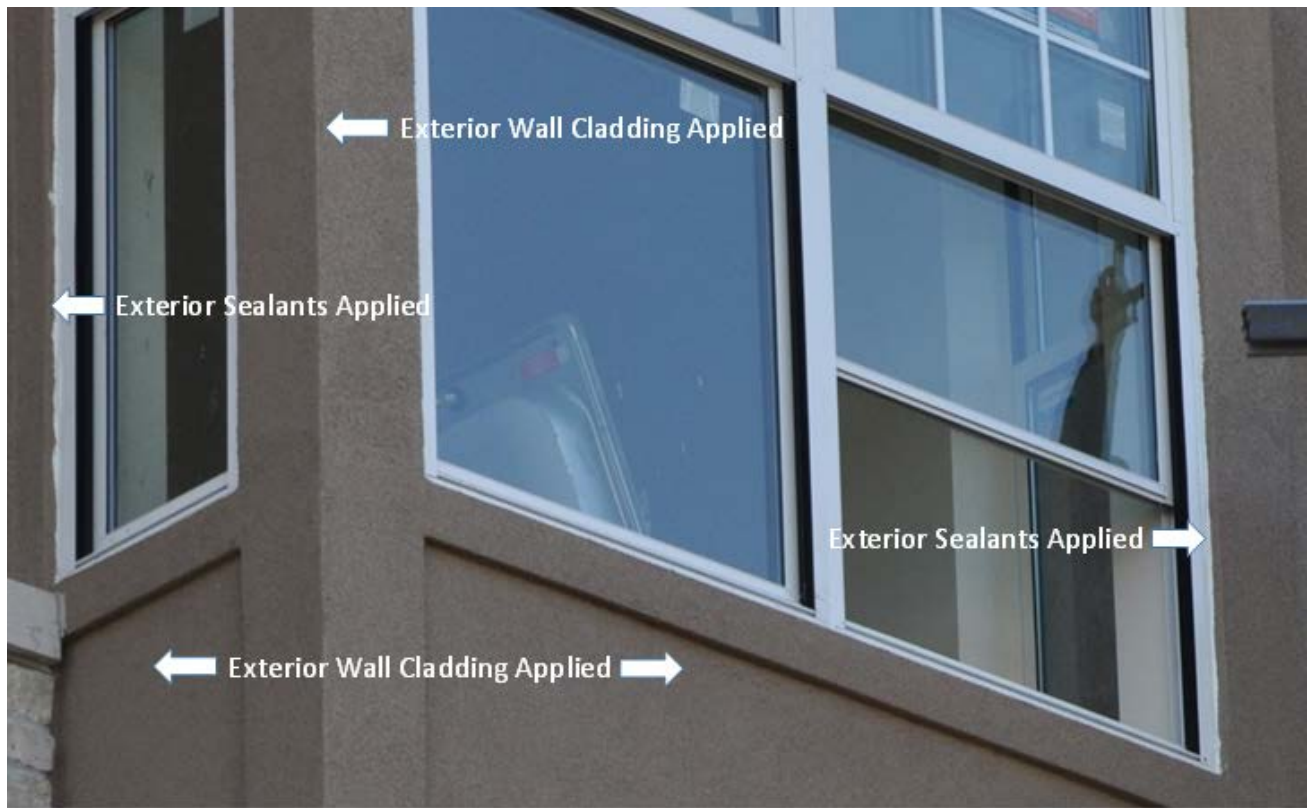


FIGURE 2: Exterior Wall Cladding and Associated Components Applied

6.4.3.1 If it is desired to test the air and vapor barrier and seals adjacent to the fenestration product prior to the exterior wall cladding being installed, the window perimeters shall be protected during that testing and the fenestration product shall not be evaluated. The same protection could be used to simulate the cladding and perimeter seals, should subsequent testing of the fenestration product be desired.

6.4.4 After the specimen(s) locations have been selected, the owner's representative shall direct the responsible contractor and manufacturer representative to remove interior finishes (if necessary), clean the specimen(s), check for proper operation, and to assure that the specimen is installed in accordance with the fenestration manufacturer's installation instructions. Care shall be taken not to disturb the interior side air seal, if present. Interior or exterior components, sealants, etc. that are required for product performance shall not be removed as some product installations require an interior air seal to perform as designed. A means to view, observe, or otherwise identify test-related water penetration beyond the planes of water penetration, as shown in Figures 3 and 4, shall be accommodated if desired by any responsible party, provided that the appropriate parties are notified that repairs must be made.

The responsible contractor shall coordinate the supply of sufficient power and water per the requirements of the AAMA accredited independent laboratory or field test agency.

The designated specimen(s) shall be inspected by the testing agency. The pretest inspection shall include the checking and recording of the following in the test report:

- The plumb, level and square of the specimen.
- The operation of the specimen.
- The designed drainage path of the specimen.
- The pre-test condition of all surfaces to confirm that they were in a condition where a judgment can be made as to whether water penetration is a result of this testing.

The fenestration manufacturer and the installer shall also be afforded the opportunity to perform an inspection prior to testing. The manufacturer and the installer shall be allowed to correct any deficiencies prior to testing. If the deficiencies cannot be corrected another sample shall be selected.

If the selected test locations are not easily accessible from both the inside and outside, significant additional cost may arise for scaffolding, swing stage, electrical and water sources. If identical products are installed at more accessible locations, these locations shall be selected for testing. Whenever possible, the architect/specifier shall define the field-testing requirements in the project specifications, including the number of tests and their location. This will provide some assurance that consideration has been given to the complexity of the field testing prior to bidding, and further communicate the requirements to those who will be responsible for, or involved in, the field testing.

6.4.5 Upon completion of the testing, the designated responsible contractor shall replace or repair items removed as specified in Sections 5.2.1, 6.4.4 and 7.5.9.

6.5 During the performance of the ASTM E1105 portion of this test, water will be sprayed on the building exterior. The responsible contractor shall take all measures necessary to protect adjacent areas from water damage that may result from unintended water exposure.

7.0 TEST PROCEDURES

7.1 Air leakage resistance and water penetration resistance tests shall be performed at pressures specified in Sections 7.4.3 and 7.4.4 unless otherwise stipulated in the Short Form Field Testing Specification. Where both tests are to be conducted in sequence, the test for air leakage resistance shall be conducted before the test for water penetration resistance.

7.1.1 The testing agency shall issue a field test plan to the entity or party contracting the test agency prior to testing that summarizes a specification review, test methods/standards, failure criteria, identifies products to be tested, number of products to be tested, locations, etc.

7.1.2 If there is reason to believe that residual water from rain or other sources may be present in the specimen, a two-minute negative (outward) pressure test followed by a two-minute positive (inward) pressure test shall be conducted at the same pressure differential used for the performance test to purge the specimen of any residual water. The specimen gaskets or weatherstrips shall be examined and shall be dry before proceeding with the air leakage resistance test.

7.2 Other than as specified in Section 7.4.2, test chambers and their associated components shall not be supported at any point by the test specimen.

7.2.1 Consideration shall be given prior to the application of water to assure that the desired loading is achieved. This is particularly critical when test samples are mounted in cavity wall applications.

7.2.2 Consideration shall be given to placement of pressure taps in relationship to the blower inlet of the chamber, as well as mullions, meeting rails/stiles, and other joints. Pressure taps should generally be placed near the center of the product.

7.3 Composite units shall be tested as a complete unit.

7.4 Air Leakage Rate

7.4.1 Air leakage rate shall be determined and adjusted to standard conditions per ASTM E783.

NOTE 5: Flexible materials are sometimes used in the construction of the test chamber. Care should be taken to stabilize the differential chamber pressure prior to recording all measurements. Large chambers in particular require sufficient time for volume adjustment to occur prior to recording the measurement.

7.4.2 Extraneous Air Test (only required if gross air leakage value exceeds the allowable value)

Measurement of extraneous air leakage (tare) readings require special consideration on many on-site installations. A review of installation details to identify sources of tare leakage shall be conducted prior to test commencement. Precautions shall be taken to minimize the extraneous air leakage. (Refer to Section 11 of ASTM E783). If the extraneous air leakage exceeds the limits specified in ASTM E783, then the chamber may be attached to the perimeter frame of the test specimen for the purposes of obtaining an accurate tare reading.

7.4.3 An air leakage resistance test shall be conducted at a minimum uniform static test pressure of 75 Pa (1.57 psf) or as specified for the project, but not to exceed 300 Pa (6.27 psf). If changes in outdoor wind speed and/or indoor air pressure are sufficient to cause static air pressure difference readings to fluctuate more than $\pm 10\%$ from the specified test pressure, one of the following methods shall be used:

- Method A: The specimen and/or test chamber shall be shielded or temporarily enclosed to minimize these fluctuations.
- Method B: Record 10 readings of air flow, evenly distributed over a 30-minute period for each of the total and tare leakage measurements. Report both the average of the ten readings and standard deviation in the test report.
- Method C: Equipment used to maintain the pressure differential during testing shall continuously monitor test pressure and self-adjust to maintain the appropriate test conditions.

7.4.4 Unless otherwise specified (see Specifier Note 3), allowable rates of air leakage for the specimen shall be 1.5 times the applicable AAMA/WDMA/CSA 101/I.S.2/A440 rate for the Product Type and Performance Class, rounded to one decimal place. Examples of allowable air leakage rates are shown in Table 1:

Product Designation	AAMA/WDMA/CSA 101/I.S.2/A440		AAMA 502	
	Allowable Air Leakage		Allowable Air Leakage	
	Test Pressure	Maximum Rate	Test Pressure	Maximum Rate
Class R-PG15- H	75 Pa (~1.57 psf)	1.5 L/s•m ² (~0.30 cfm/ft ²)	75 Pa (~1.57 psf)	2.3 L/s•m ² (~0.45 cfm/ft ²)
Class LC-PG25-SD	75 Pa (~1.57 psf)	1.5 L/s•m ² (~0.30 cfm/ft ²)	75 Pa (~1.57 psf)	2.3 L/s•m ² (~0.45 cfm/ft ²)
Class CW-PG30-C	75 Pa (~1.57 psf)	1.5 L/s•m ² (~0.30 cfm/ft ²)	75 Pa (~1.57 psf)	2.3 L/s•m ² (~0.45 cfm/ft ²)
Class AW-PG40-FW	300 Pa (~6.27 psf)	0.5 L/s•m ² (~0.10 cfm/ft ²)	300 Pa (~6.27 psf)	0.8 L/s•m ² (~0.15 cfm/ft ²)

TABLE 1: Examples of Allowable Air Leakage Rates

When field air leakage tests are performed on project specific size fenestration products that are smaller than the maximum size tested and documented in the laboratory test report, the acceptable air infiltration rate for these smaller sized products shall be permitted to be increased, at the discretion of the specifier and manufacturer. It is highly recommended that the allowable leakage rate be established prior to completion of the project specification.

This is especially critical for products where the junctions at the meeting rails, corners and/or hardware attachments are, for example, the primary source of air leakage. Since allowable leakage is calculated based on product area, a smaller size product may be penalized since it employs the same design as a larger product that complies with AAMA/WDMA/CSA 101/I.S.2/A440.

NOTE 6: The purpose of in-field air leakage testing is to provide reasonable assurance that the fenestration product will perform to its specified or rated level with allowances given for field reductions. The allowable air leakage used for tests conducted in the field, shown in Table 1, are not the same as the laboratory allowance. This is to allow for field conditions and test methods that vary from the laboratory test conditions and test methods. These conditions are primarily related to the ambient environmental conditions and the installation. The rated product performance is based on laboratory testing performed under controlled laboratory conditions. The temperature, wind and barometric pressure conditions during a field test will typically vary from the standard laboratory conditions.

The field installation conditions also influence the product performance. Products tested in the laboratory are perfectly plumb, level and square in a precision opening. Field test specimens, although installed within acceptable industry tolerances, are rarely perfectly plumb, level and square. Shipping, handling, acts of subsequent trades, aging and other environmental conditions all may have an adverse effect upon the performance of the installed specimen. A 50% increase in the allowable air leakage for field testing is specified as a reasonable adjustment for the differences between a laboratory test environment and a field test environment.

NOTE 7: The IP equivalents identified are for approximate reference only and do not in any way imply accuracy of the measurement or the equipment. Precision and bias statements are provided in the applicable test methods referenced in this Test Method

NOTE 8: The information is for example only and is not a complete list of all fenestration product designations. Refer to Table 25 in AAMA/WDMA/CSA 101/I.S.2/A440-05, Table 27 in AAMA/WDMA/CSA 101/I.S.2/A440-08, or Table 12.2 in AAMA/WDMA/CSA 101/I.S.2/A440-11 and -17 for a complete listing of all designations and maximum allowable air leakage rates.

7.5 Water Penetration Resistance

7.5.1 Water penetration resistance performance shall be determined per ASTM E1105 using Procedure B “cyclic static air pressure difference,” except for AW windows which shall be tested using Procedure A, “uniform static air pressure difference.” The test pressure shall not be less than 91 Pa (1.90 psf).

7.5.2 The ASTM E1105 test consists of sealing a chamber to the surrounding conditions adjacent to the specimen to be tested and supplying or exhausting air to the chamber at the rate required to maintain the desired air pressure difference across the specimen. Simultaneous to the application of air pressure difference, water shall be applied to the exterior face of the specimen at the required rate, per ASTM E 1105, while observing for any water penetration at the interior which is identifiable as originating from the specimen.

7.5.3 Unless otherwise specified in field testing section of the contract documents (see Specifier Note 4) water penetration resistance tests shall be conducted at a static test pressure equal to 2/3 of the tested and rated laboratory performance test pressure as indicated by the applicable product designation in AAMA/WDMA/CSA 101/I.S. 2/A440. For example, a product tested or rated as H-CW50 shall be field tested at a pressure differential of $0.667 \times 360 \text{ Pa}$ (7.50 psf) = 240 Pa (5.00 psf).

NOTE 9: *The default pressures used for water penetration resistance tests conducted in the field are not the same as the laboratory test pressure. This is to allow for field conditions and test methods that vary from the laboratory test conditions and test methods. These conditions are primarily related to the ambient environmental conditions and the installation. The rated product performance is based on laboratory testing performed under controlled laboratory conditions. The temperature, wind and barometric pressure conditions during a field test will typically vary from the standard laboratory conditions.*

The field installation conditions also influence the product performance. Products tested in the laboratory are perfectly plumb, level and square in a precision opening. Field test specimens, although installed within acceptable industry tolerances, are rarely perfectly plumb, level and square. Shipping, handling, acts of subsequent trades, aging and other environmental conditions all may have an adverse effect upon the performance of the installed specimen. A 1/3 reduction of the test pressure for field testing is specified as a reasonable adjustment for the differences between a laboratory test environment and a field test environment.

7.5.4 Prior to each test, a visual confirmation shall be made that all nozzles are flowing freely and that the straightness and angularity of the nozzles is normal to the plane of the test specimen. Any non-functioning nozzles shall be repaired or replaced.

7.5.5 During the test period, a laboratory or test agency representative shall visually confirm that all test surfaces are uniformly and sufficiently wetted. For test samples larger than the calibrated rack, multiple spray racks may be used provided that the calibration for each rack has been performed as described previously and a sufficient water source is used to supply the pressure established during calibration as recorded in Section 8.2.

7.5.6 In the event the water source is not sufficient to supply the entire test area at the calibrated pressure it is permissible to test smaller sections of the test area with smaller water spray racks. The test report shall note this exception and any influence on the test results. The specimen shall be allowed to drain for at least 15 minutes between tests. At the commencement of the next test, there shall be pressure applied for five minutes without the application of water to ensure there is no residual water remaining in the product that could affect the test results. If residual water is observed after the application of pressure, then additional drainage time shall be allowed prior to commencement of the next test.

For products segmented into multiple test areas due to size, the evaluation of water penetration shall be confined to the respective area subject to testing.

7.5.7 With all operable portions of the specimen closed and locked, the specimen shall be subjected to a water penetration test in accordance with ASTM E1105 procedure “A” or “B” in accordance with Section 7.5.1. Procedure “A” shall consist of a 15-minute test with continuous pressure and water application. When using procedure “B” each of the four cycles shall consist of five minutes with pressure applied and one minute with the pressure released during which the water spray is continuously applied.

7.5.7.1 When testing side-hinged door and dual-action side-hinged door systems, water penetration through the locking/latching hardware shall be permitted to be excluded, provided that the exclusion is clearly identified in the test report. This shall include a complete description of the exclusion method employed. If it is desired to test a product including the locking/latching hardware, the water penetration test shall be conducted with the exterior side of the locking/latching hardware fully exposed to the water spray with no exclusion methods employed. In this case no exclusion statement shall be required.

NOTE 10: *Side-hinged door locking/latching hardware is generally an aftermarket installation and is therefore beyond the control of the door system manufacturer.*

7.5.7.2 During the test period the water source provided to the spray rack shall be regulated to the same pressure used for calibration of the spray rack. The distance from the spray rack to the test specimen shall be within +/- 50 mm (+/- 2 in) of the spacing used for calibration. In addition, the pressure gauge shall be located at the same position in the spray rack that was used for calibration.

7.5.7.3 If the water source is not capable of achieving calibration pressure, appropriate steps shall be taken to increase the source pressure. If the water pressure cannot be increased to the pressure used during calibration it shall be noted in the test report along with the reason that the water pressure could not be achieved, and that testing using a water pressure lower than the calibrated water pressure is a deviation from the ASTM E1105 procedure.

7.5.7.4 If the rack to specimen spacing or pressure gauge location cannot be duplicated, a recalibration of the rack in accordance with ASTM E 1105 is required.

7.5.8 Observe and note all points of water penetration, if any, that occur during the test. If the origin of the water penetration cannot be definitively attributed to either the fenestration product specimen or the joint between the fenestration product specimen and the surrounding condition, a forensic evaluation shall be performed using the procedures outlined in AAMA 511, while maintaining the test pressures defined in the field-testing specifications and employing the test methods defined in AAMA 502.

7.5.8.1 Water penetration shall be defined as follows:

- **Water penetration attributable to the surrounding wall conditions:** the presence of water not contained within drained flashing, gutters, and sills which did not originate from the fenestration product or the joint between the fenestration product specimen and the wall/roof.
- **Water penetration attributable to the fenestration product specimen:** shall be per ASTM E1105,
- **Water penetration attributable to the perimeter joint:** water not controlled by a water management system that indisputably originates at the perimeter joint (see Figures 3 and 4).

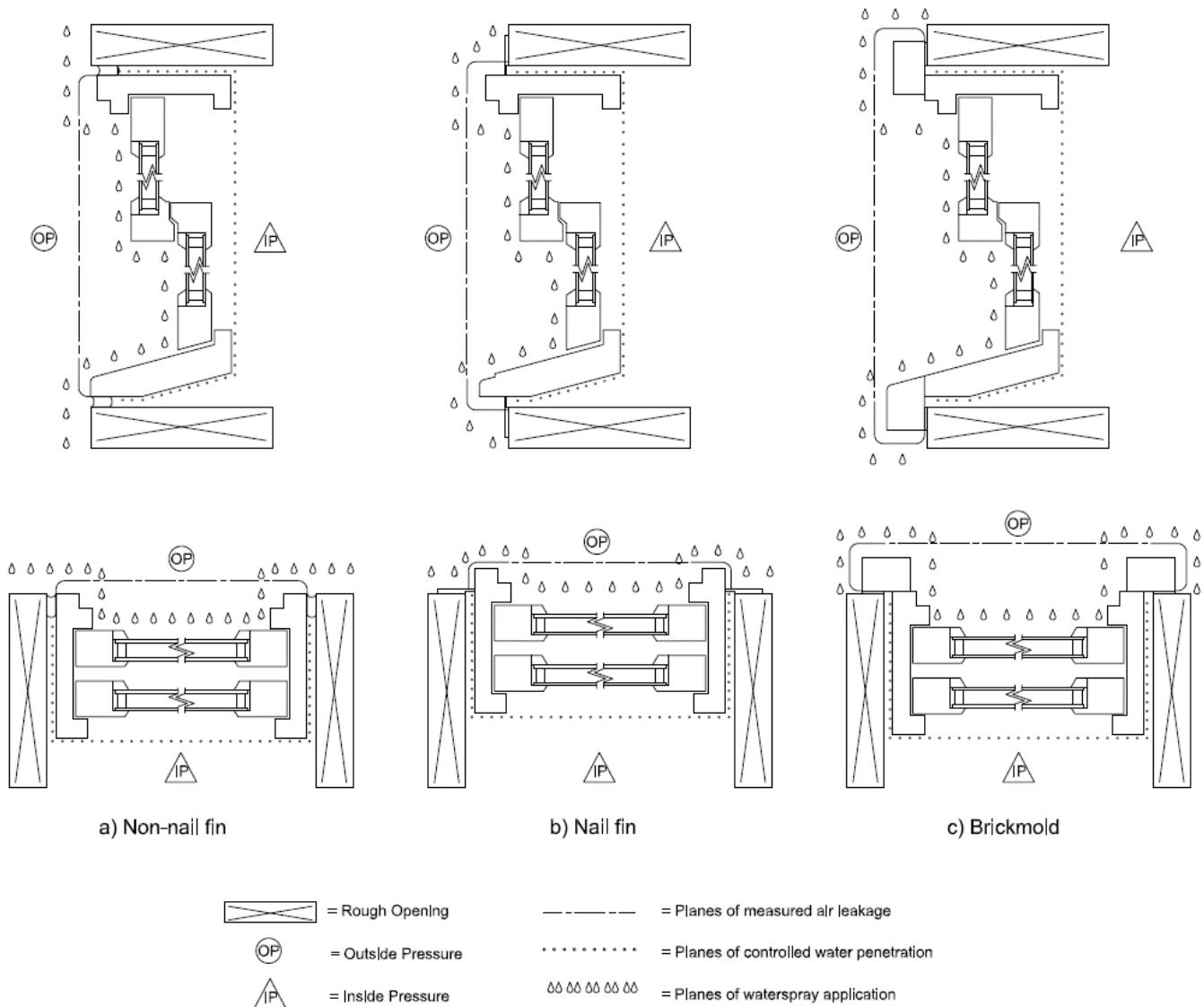


FIGURE 3: Plane of Water Penetration

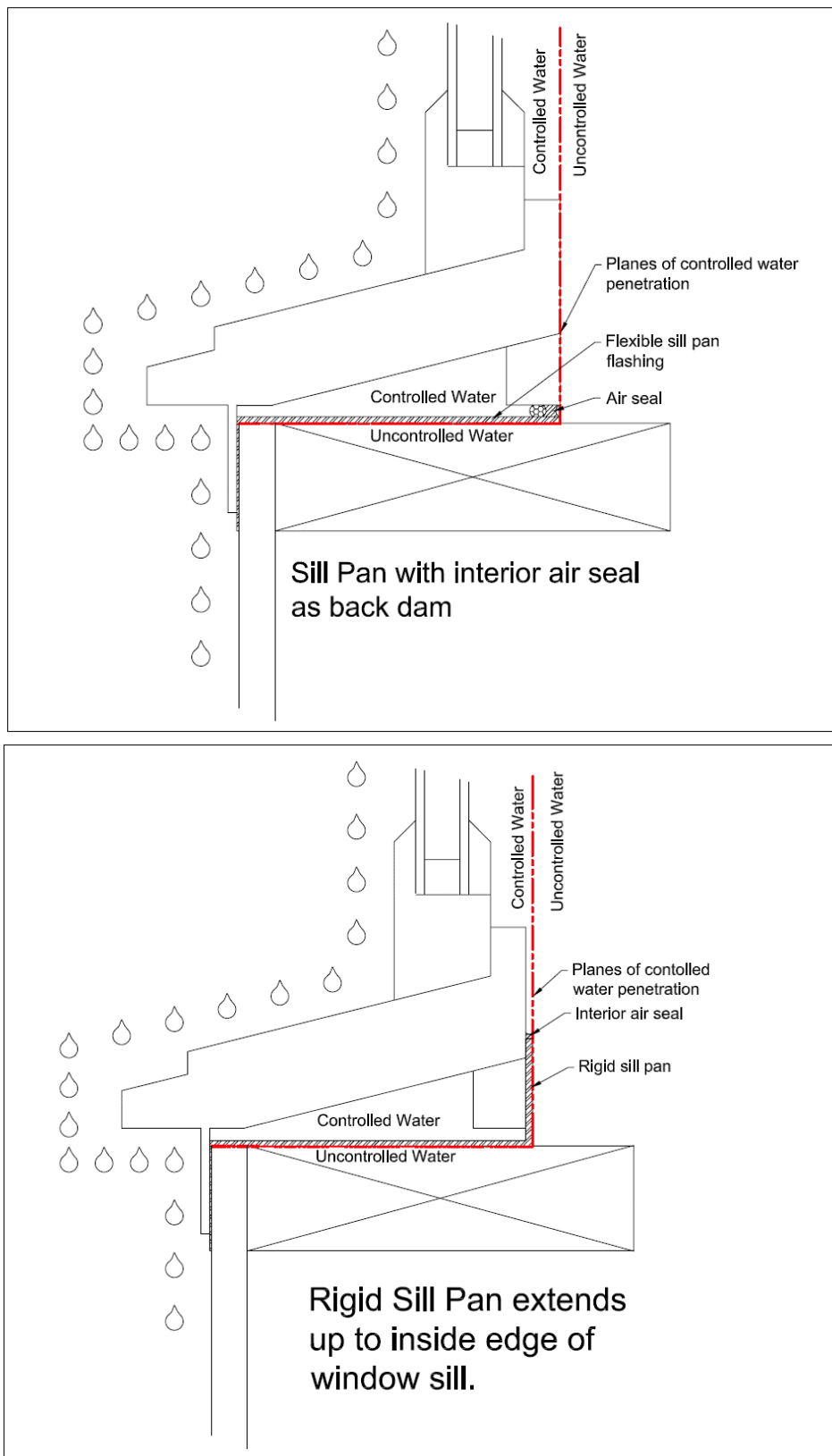


FIGURE 4: Planes of Water Penetration when incorporating a Flexible or Rigid Sill Pan

7.5.9 If any of the specimens do not conform to the prescribed air leakage resistance and water penetration resistance requirements, the manufacturer and/or the installer shall be afforded the opportunity to perform a site inspection and determine the reason for non-compliance. Non-compliant specimens shall be repaired as required and retested. The remedial work shall be recorded and approved by the specifying authority, architect and/or owner. Upon satisfactorily passing a retest, the remedial work performed shall become punch list items to randomly check for similar conditions on the remainder of the project. When the inspection and remediation of a specimen requires removal and re-installation of parts or the whole specimen, it shall be at the sole discretion of the lab/testing agency to determine the feasibility of re-testing the same specimen.

8.0 CALIBRATION AND VALIDATION

8.1 At least every six (6) months, the pressure measuring apparatus shall be calibrated to measure the test pressure difference within a tolerance of $\pm 2\%$ or ± 0.01 in. of water column (± 2.5 Pa), whichever is greater.

8.2 Guidelines for Validation of Air Infiltration Equipment Used for On-Site Testing

8.2.1 A full validation shall be conducted under controlled conditions in accordance with the tolerances in ASTM E283/E283M Section 9 for each air flow measurement system used for on-site testing and utilize the validation apparatus described in Figure 1 of this document. Controlled validation shall be performed at least every six (6) months.

NOTE 11: *ASTM E283/E283M is referenced because validation requirements in ASTM E783 were still under development at the time this version of AAMA 502 was being worked on.*

8.2.2 The full validation shall consist of a minimum of three flow standards (verification plates) between 7.5 CFM and at least 20% of the equipment capacity using the validation apparatus described in Figure 1.

8.2.3 Onsite Spot Check: To ensure there is no change in measurement accuracy as a result of transport, a spot check shall be made at the beginning of each day of testing as follows:

8.2.3.1 Attach the air flow metering system and identical piping scheme to be used for testing to the validation apparatus shown in Figure 5. Determine the measured airflow through flow standards of 7.5 CFM and a point within 20% of the equipment capacity (verification plates) at the specified pressure for the flow standards.

8.2.3.2 If the results for both verification standards are within the allowable tolerance for accuracy as defined in ASTM E783, testing may proceed.

8.2.3.3 If either of the results are not within the allowable tolerance of accuracy established in ASTM E783, identify and correct the cause of the discrepancy. Verify the discrepancy has been corrected by completing a full verification in accordance with Section 8.2.1.

8.2.4 Records of controlled validation and onsite spot-checks shall be maintained for a minimum of four (4) years. Records of the most current validation and onsite spot check shall be made available for inspection at the jobsite prior to the commencement of testing. A statement that the spot check(s) was conducted was conducted prior to testing shall be part of the test report.

8.3 Guidelines for Calibration of Water Spray Equipment Used for On-Site Testing

8.3.1 A full calibration shall be conducted under controlled conditions in accordance with ASTM E1105 on the maximum height and width spray rack. Controlled calibration shall be performed at least every six (6) months.

8.3.2 The water source pressure shall be recorded during calibration using a calibrated pressure gauge that will remain a part of the spray rack throughout the six-month calibration period.

8.3.3 The spacing between the nozzles and the front plane of the catch box shall be recorded as part of the calibration data.

8.3.4 Records of controlled calibration shall be maintained for a minimum of four (4) years. Records of the most current calibrations shall be made available upon request for inspection at the jobsite prior to the commencement of testing.

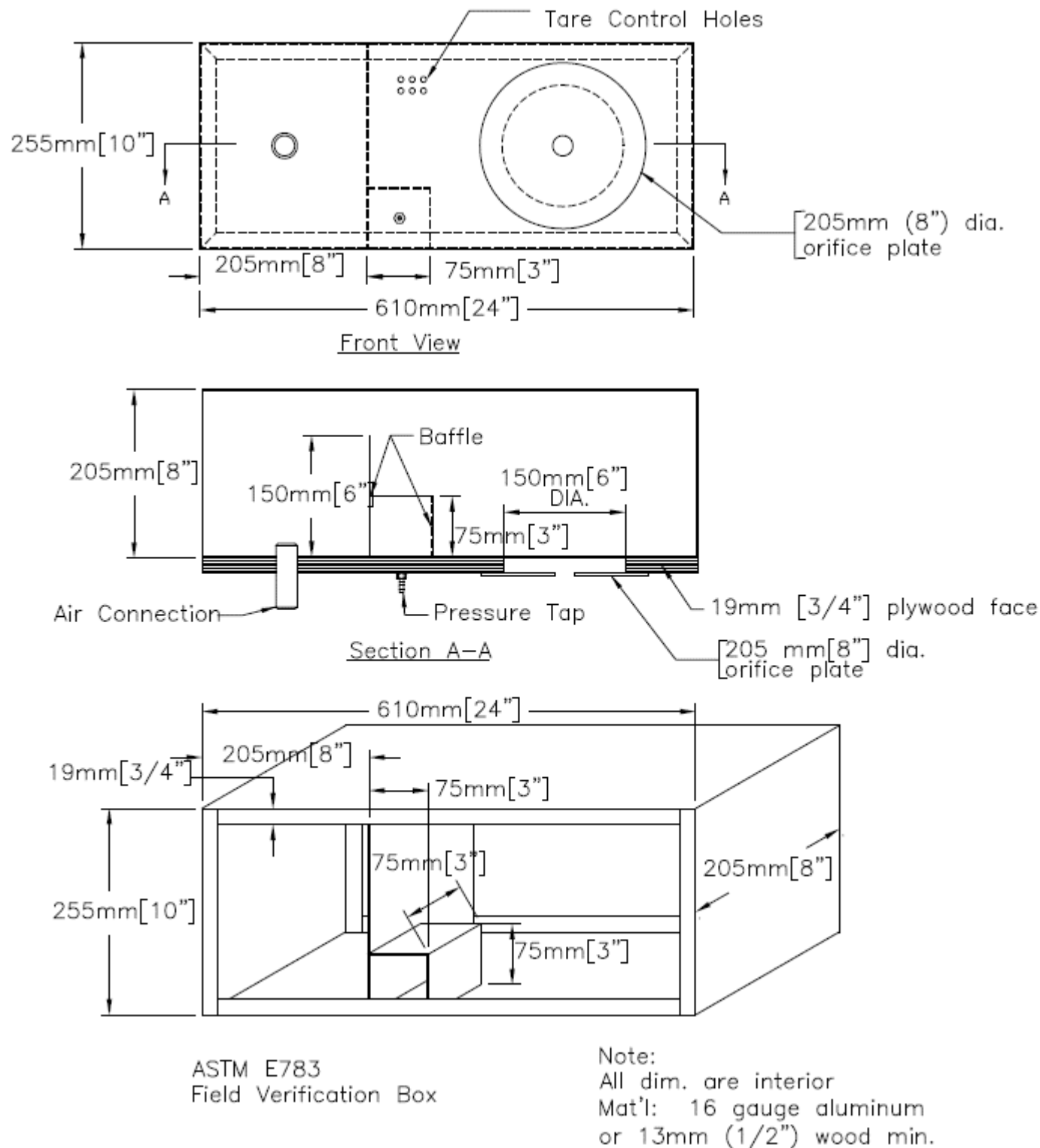


FIGURE 5: Validation Apparatus

9.0 TEST REPORTS

9.1 The report shall include enough information to reproduce the test. The following information shall be included, at a minimum:

9.1.1 General

- The testing agency;
- Name of the individual(s) performing the tests and the FenestrationMaster® or FenestrationAssociate® in responsible charge;
- Name, title, and company affiliation of those witnessing all or part of the test(s);
NOTE 12: This list is meant to include individuals with significant interest in the test or official observers.
- Date and time of test, date of report, identification and location of the building shall be identified;
- The date of the last equipment calibration and the location of calibration records shall be included in the report and;
- If testing per ASTM E783 is performed, a statement that the spot check(s) required by Section 8.2.3 were conducted prior to testing.

9.1.2 Fenestration Product Description

- The manufacturer, model, operation type, dimensions, materials, etc.;
- Identification and location of specimen(s) within the building;
- Physical condition of all elements of the specimen including the weep or water management system;
- Description of any modifications made to the specimen material to the test;
- Number of retests, etc.;
- The plumb, level and square condition of the tested specimen;
- A statement documenting the inspection of the wall construction and other conditions adjacent to the test specimen;
- Other details relevant to the installation of the tested specimen and;
- Photographs of interior and exterior views of the test specimen, and any other details that could affect the test results.

9.1.3 Sampling Procedures

If applicable, describe or list the procedures established from Section 6.0.

9.1.4 Test Parameters

Record all the following that apply to the application of the test(s):

- Reference to project specification(s) and test standards used to determine procedures, test pressure(s), and pass/fail criteria;
- List or describe the specified cyclic and/or static pressure differential(s) used in the test;
- Chamber description, including whether the chamber was affixed to the interior or exterior of the wall/roof, and a detailed description (to include sketches or photos showing location, if appropriate) of the chamber attachment to the specimen;
- A description of any trim or wall cladding, or other materials removed during the test to provide a path for observation or to facilitate chamber attachment;
- A description of relative ambient conditions, and any measures that were taken to control ambient conditions;
- Clearly identify any elements of the specimen that were not tested;
- Identify all intended areas or elements to be isolated, including techniques utilized;
- A list of any changes, modifications or adjustments made to test specimens either prior to or during the testing sequence and ;
 - Document any adjustments/repairs made after a failing test and prior to retesting.
- A statement confirming that:
 - The sample was inspected immediately prior to the test or installation of the chamber if it conceals portions of the window,
 - The original conditions were observed and documented
 - All surfaces were dry, such that water observed during or after testing was produced by the test itself and no other possible source.

9.1.5 Test Results

Record all the following:

- Actual and allowable air leakage for the window or door;
- Actual water penetration for the window or door;
- Actual water penetration for the perimeter condition;
- Environmental conditions as measured at the time of the test, including:
 - o Wind, speed and direction
 - o Precipitation
 - o Barometric pressure
 - o Ambient temperature
- Photos that support the description of the test specimen and the test results.

9.1.6 Compliance Statement

Make a statement that the tests were conducted in accordance with this specification or completely describe any deviation, including the purpose and the technical rationale for the deviation.

Changes from AAMA 502-12 to AAMA 502-21

- Various editorial changes were made.
- This edition was a major revision from the previous edition. Several sections were added to meet current FGIA publication requirements, while other sections were relocated.
- Short Form Specification- changed the minimum number of specimens to be tested to “1% of the fenestration product specimens but no fewer than three...”
- 0.0 Introduction- new section
- 1.0 Scope- added section 1.3
- 2.0 Significance and Use- new section
- 4.0 Definitions- new section
 - Language in 502-12 Section 4.2 was moved to Section 7
- 6.0 Sampling, Site Preparation, and Responsible Parties
 - Note 3: Removed reference to AAMA 204 and added reference to LAP-3
 - 6.4: revised to allow inspection and correction of issues with product by manufacturers and installers prior to testing; installation of exterior façade must be complete prior to testing
- 7.0 Test Procedures
 - 7.1.1: test agency must issue a test plan prior to testing
 - 7.4, 7.5: Now includes language on air leakage and water penetration testing previously found in LAP-3-19
 - 7.5.8.1: revised descriptions of water penetration
 - Figures 3 and 4: show the planes of controlled and uncontrolled water penetration for various installation types and flashing conditions
- 8.0 Calibration and Validation
 - Now includes language on calibration and validation of test equipment previously found in LAP-3-19
- 9.0 Test Reports
 - Added items to the minimum requirements for test reports
- This document was updated by the Field Test and Forensic Investigation Document Update Task Group

Changes from AAMA 502-11 to AAMA 502-12

- Various editorial changes were made
- Added new Section 1.4
- Updated Table 1 to reference Product Designations in AAMA/WDMA/CSA 101/I.S.2/A440-11 (addition of CW Class, removal of C and HC Classes)
- Added new NOTE 4
- Updated NOTE 5 reference air infiltration rates for smaller sized products “shall be permitted” to be increased “at the discretion of the specifier and manufacturer.”

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